So the main advantage of **public key cryptography** over private key cryptography is that it allows for secure communication without the need for the sender and receiver to first share a secret key, which could be intercepted by a third party.

- ***Bitcoin*** transactions involve transferring a certain amount of ***bitcoin*** **directly** to another person's account.
    - You wouldn't want someone to just steal the money your friend sent via Venmo because the payment rail is insecure, would you?

- Encryption is a way to keep important information guarded from malicious actors as it travels through the network, some of whom, such as hackers, might deviate the funds to their account instead.

- As an additional measure of protection, to keep the **transaction details** safe and secure in ***Bitcoin***, a ***UNIQUE*** signature is added to each transaction. This signature acts like a secret code that ensures that no one can change any part of the transaction without the software detecting it and marking it as invalid.

Alice [Cash Account]

Debit	Credit
	$100

Bob [Cash Account]

Debit	Credit
$100	

Public Ledger (Digital Receipt)

Alice	Bob
$100	$100
Out	In
Signature	Signature

Comparing Bitcoin Transactions to Traditional Banking

- In traditional banking, a **PIN** is used to authenticate transactions, similar to how a **private key** is used to sign transactions in *blockchains*.

A simple analogy for this process would be a person accessing their bank account number with a private pin (**private key**), and then using their own personal signature (unique signature) on an online check (***digital currency***) to send money (**make a transaction**) to someone else (another user). Just like how a person's signature on a check verifies their identity and authorizes the transaction, the digital signature using a **private key** verifies the identity and authorizes the transaction of digital currency.